

BenGER Transformation: Exploring Automated Rubric Generation to Reduce LLM Grading Variance in German Legal Exams

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Large language models (LLMs) can grade answers to German law exams at scale. But the same answer gets different scores from different models, and even from one model on a second run. Some examiners grade with a grading sheet (*Bewertungsbogen*) that splits the expected solution into small steps worth fixed points. We ask whether a sheet that a model writes from the case and its reference solution makes LLM grades steadier, and at what cost to agreement with human graders. A first study on 15 exams and 45 human-graded answers used sheets from 12 models. On average the sheets did not make the stronger grading models (judges) steadier and tended to lower agreement with the human graders. With a sheet, several judges gave high scores to empty answers or to answers that only copy the case. In the pipeline we present, one model writes small steps with explicit requirements, including paths for defensible alternative solutions and auxiliary opinions (*Hilfsgutachten*). The judge must back every point with a quote, and a fixed program checks that each quote stands in the answer. On the scripts of one police-law exam, graded on its author's own sheet, we test whether sheets make grades steadier and closer to the examiner's, which models write and use sheets well, and whether the pipeline meets German examination law. First runs on 15 scripts suggest that a sheet does not make repeated runs steadier, but brings the grades much closer to the examiner's. [PENDING: results of the full benchmark and the recommended pipeline]

CCS Concepts: • **Applied computing** → Law; Education; • **Computing methodologies** → Natural language processing.

Additional Key Words and Phrases: LLM-as-a-judge, grading rubrics, legal education, German law examinations, automated assessment, reliability and validity

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1 Introduction

German law students learn and are examined through case solutions (*Falllösungen*). A candidate receives a fact pattern (*Sachverhalt*) and writes a legal opinion in a prescribed appraisal style (*Gutachtenstil*), which the state examinations grade on a statutory scale from 0 to 18 grade points [6]. Different examiners often give the same answer different grades [31]. Some correctors therefore first note what each part of the expected solution is worth [23, p. 330], and a few examiners make this binding in a grading sheet (*Bewertungsbogen*) [29, 61]. Such a sheet splits the expected solution into small steps, each worth some assessment units (*Bewertungseinheiten*, BE), and a key maps the BE total to a grade. Appendix A lists the German and the technical terms we use.

Large language models (LLMs) can grade such texts at scale; an LLM used as a grader is called a *judge*. On German case solutions, a judge that applies the same 10 general criteria to every exam stays within the range of human disagreement, but its scores vary between judges and between

repeated runs [50]. For candidates this is a matter of equal opportunity (*Chancengleichheit*), since comparable candidates must be graded by comparable standards [7, p. 52]. We test the examiners' remedy: one model, the *generator*, writes an exam-specific grading sheet from the case and its reference solution (*Musterlösung*), and the judge grades against it. The sheet also gives the examiner a document to read and correct.

Study 1 tested this remedy on 15 exams with 45 answers and blind human grades. On average the sheets did not make the stronger judges steadier and tended to lower agreement with the human graders. With a sheet, several judges also gave high scores to empty answers or to answers that only copy the case (Section 6.1). The pipeline we present (Section 4) is designed against these failures. Study 2 tests it on the written answers (*scripts*) to a police-law exam set by one of us and graded on the author's own sheet and key, and it asks what German examination law requires of such grading. Four research questions guide the paper:

- **RQ1 Variance.** Does a generated sheet make LLM grades steadier than grading without one, counting the scatter between runs, judges and the sheets one generator may write? Does it beat averaging several runs at equal cost?
- **RQ2 Validity.** How close are sheet-based grades to the examiner's and to a second human grader, and how many students would they wrongly pass or fail?
- **RQ3 Generators and judges.** Which models write sheets that yield grades close to the examiner's at low cost, and which judges use them well?
- **RQ4 Law.** Does sheet-based grading meet what German examination law asks of grading, on defensible (*vertretbar*) alternative solutions, follow-on errors (*Folgefehler*), reasons for each point and equal opportunity, and which uses does the evidence support?

Our contributions are:

1. A pipeline that turns doctrine into checkable structure (Section 4): steps with explicit requirements, separate paths for defensible alternative solutions, owed auxiliary opinions (*Hilfsgutachten*), and points only for quotes that a fixed program finds in the answer.
2. A study of all scripts of one exam that splits the scatter of grades into its sources (runs, judges, sheets) and compares sheets with averaging runs at equal cost (RQ1).
3. Agreement with the examiner, a second human grader and the examiner's own second grading, counting the students an LLM would wrongly pass or fail (RQ2), and a small benchmark of generators and judges (RQ3).
4. A legal analysis, co-written with the exam's author, that maps each requirement of examination law to a design feature and a test (Section 2, RQ4).

[PENDING: one plain sentence per research question once E1 to E5 have run]

Most analyses are exploratory; one test is fixed in advance and will be published before the examiner grades the 23 held-back scripts. [PENDING: OSF registration link]

2 Grading practice in German law exams

[DRAFT: to be co-written with the exam author; every pending item below needs the author's input or experiments E2, E4 and E5]

The setting. The statutory scale has 7 grades (*Notenstufen*) from 0 to 18 grade points, and 4 points (*ausreichend*, “sufficient”) is the lowest grade that still meets average requirements despite its flaws [6, § 1]. In practice a single exam is passed from this grade, although the rules on passing are state (*Land*) law. Examiners set their own standards for an exam and must apply them to every script alike [9]. University exams usually have one grader [61, p. 29]; some correctors first build a private point scheme [23, p. 330], and a few examiners make it binding in a sheet [29], but such sheets are rare [61, p. 3]. Two principles bind any sheet. A defensible solution (*vertretbare Lösung*), consistently argued with weighty reasons, must not be marked as wrong, which for exams

that control access to a profession follows from Art. 12(1) GG [7, p. 55]. And a follow-on error (*Folgefehler*) should cost points only once, so a later step that correctly builds on the candidate's own earlier decision still earns credit. [PENDING: exam author: source for the follow-on-error principle]

Three paradigms. We compare three ways of grading with our sheet (Table 7 in Appendix B). The exam author's sheet (D2) is a list of steps with absolute BE, grouped into sections. Gregor Roth's sheet for a civil-law exam (Z2) is a tree in which each item carries a percentage of the item above it. Each of the 40 smallest items gets a grade from 0 to 18, and the final grade is their average weighted by these percentages, so the method needs no key [61, p. 12]. A third practice is common as well: about 80% of the BE go to steps, and the grader distributes the remaining 20% freely, for instance for structure or the overall impression. We do not adopt it, because the free share is tied to nothing in the answer that a second grader or a judge could check, so it invites variance and cannot be reproduced. Our sheet follows the first paradigm: every BE, in half points, belongs to a step, each step lists the requirements that earn it, and code does all arithmetic (Section 4).

Q1: How should the BE be distributed? The author's sheet spreads 100 BE over 46 steps with at most 10 BE per step, and some step weights changed between years (Section 5.2) [PENDING: exam author: why]. We took the step-size guidance of our generator (0.5 to 4 BE per step, at most 10 BE) from the two expert sheets, so step size is an input of our design, not a finding. E3 compares the BE distributions of generated sheets with the author's sheet and E5 with Roth's. [PENDING: BE distribution of the expert sheet and of generated sheets (E3)]

Q2: What earns credit, including method? In the author's sheet every step credits an issue spotted or an argument made. No step credits a bare result, although some step titles state a sub-result. Our generator follows the same rule: a result earns credit only with the reasoning that carries it, and where several results are defensible, the requirements credit the argument [7, p. 55]. The exam author credits method (structure, appraisal style, precise language) inside each section rather than as a separate block [61, p. 14]. [PENDING: exam author: confirm for D2]

Q3: Where is the pass mark? The author's key passes a script at 40% of the BE. Where the pass mark lies is part of the examiner's own evaluation; it cannot be fixed rigidly without regard to average results [7, p. 51], and once set it applies to every script alike [9]. The mark decides which students a judge passes or fails differently from the examiner, and E2 sweeps it from 35% to 55% on stored outputs. [PENDING: E2]

Q4: Is the key linear or curved? The author's key is piecewise linear with a kink at the pass mark: below it a grade point spans 10 BE, and from the pass mark upwards 4 BE, up to grade 18 at 96 BE. The same shape, with passing costing more BE per grade than the top grades, is proposed in [61, p. 15]. [PENDING: exam author: confirm the reasons for D2] Roth's paradigm needs no key, since it averages statutory grades and passes from grade 4.

Q5: How are cardinal errors treated? A cardinal error (*Kardinalfehler*) is one that by itself can make a script fail. Examiners disagree on what counts, and the literature asks each sheet to define it in advance, with minus points or a grade cap [61, pp. 15 f.]. [PENDING: exam author: examples and expected loss for D2] Our sheet has no such cap yet; the follow-on-error rule and the standard of each requirement decide how far an early error carries (Section 4.3). E4 tests 5 fake answers with the losses the author expects.

3 Related work

Grading with LLMs. A judge reads an answer and returns a score. Such judges agree with human graders on average [43, 70] but are less dependable on a single answer. The same answer can get a different score when graded again [28, 57], scores change when answers are shown in another order [58, 64], and judges favour texts written by their own model [51]. A grader can also be

steady and still wrong [32, 68], so each use must be checked against human graders [4, 10, 26]. Testing research splits score differences into parts due to the student, the grader and the occasion (generalizability theory) [5]; this has been applied to LLM graders [60, 65], and the wording of the grading prompt has been treated as one such part [49]. Most fixes change the grader, for example by pooling several models [63]. We change the sheet the grader uses, and we count the sheet a model happens to write as a source of differences of its own.

Grading sheets written by models. A rubric is a scoring guide that states what earns credit, and a checklist breaks it into yes-or-no items. Such items can make LLM grading more consistent [11, 15, 36, 40, 69], and rubrics written for one question work better than general ones [33, 52, 62]. Human graders, too, agree more when they score an answer part by part than when they form one overall impression, at a cost in time [37]; for an LLM, the time becomes computing cost. Models can write such rubrics: they derive weighted criteria from a reference answer [22, 27, 42] and point schemes for exam questions and proofs [25, 45, 46, 67]. Generated rubrics still fall short of expert ones [66, 71], and a regenerated marking scheme graded less accurately than the one the human graders had used [45]. Because models that add up points themselves make mistakes [38, 54], our generator states only relative weights and code allocates the points. Our sheet also encodes two rules of examination law: defensible alternative solutions, and follow-on errors that cost points only once.

Evidence and gaming. Attribution research asks whether a text is backed by the sources it cites [24, 56], and some systems require exact quotes and check them in code [30, 35, 48]. Automated scoring has long been open to gaming by padding, nonsense or keyword lists [17, 55], and LLM graders can be steered by instructions hidden in an answer [12, 41]. Our judge must quote the answer for every credit, code removes a credit whose quote is missing or turned into its opposite, and fake answers test every judge before use.

Grading law exams. Research first asked whether LLMs pass law exams [14, 34, 47] and now asks whether they can grade them. The graders receive model answers, marked passages, or rubrics written by experts or exam offices [1, 13, 16, 21, 53, 54, 59], while others split answers into single legal points [18] or derive issue trees from court judgments [39]. German examiners disagree widely [31]. An examiner's own point scheme cut the spread between human correctors by almost half [61], and LLM grades have been validated against the same examiner's grades on the examiner's own scheme [2]. To our knowledge, no study on law exams has a model write the point scheme from the reference solution, with scoreable paths for defensible alternatives and a follow-on-error standard per requirement, and tests the result against the examiner's own sheet and grades.

4 The pipeline

Figure 1 shows the pipeline. Models do two things only: the generator writes the sheet and says how much each part weighs, and the judge grades an answer against the sheet and quotes the answer for every credit. Plain code, a fixed program rather than a model, does everything that can be computed or checked. Since models that add up points themselves make mistakes [38, 54], no model ever states points.

4.1 Generation: the checklist sheet

The generator receives the case text, the reference solution and fixed instructions, the same for every model (Appendix K), and returns a *checklist sheet* with these parts:

- *Sections and steps.* Each step names one expected legal performance and carries a relative weight. The exam's focal problems (*Klausurschwerpunkte*) are marked groups of steps, and each step belongs to a work result (*Arbeitsergebnis*), such as an opinion or a draft judgment.

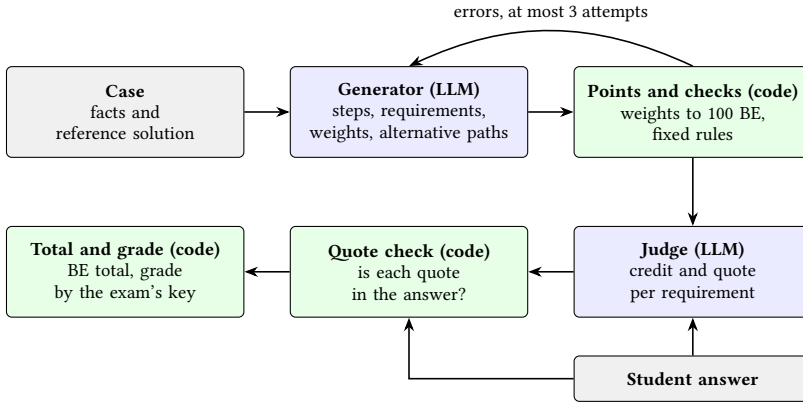


Fig. 1. The grading-sheet pipeline. Blue boxes are model steps, green boxes are code. The judge also receives the case facts and the reference solution.

- *Requirements.* Each step lists 1 to 6 requirements, each with a share of the step and a standard (*Maßstab*). A requirement is either correct (*richtig*), judged against the correct law, or consistent (*folgerichtig*), judged against the candidate’s own earlier decision; the latter carries the follow-on-error rule. Each requirement names something the answer must write, which a quote can show. None may be met by leaving something out, none may be conditional, and none may ask only for a norm, a term or a result.
- *Forks and alternative paths.* A dispute that changes only a result or its reasons stays inside one step, and its requirement credits the argument whatever the result. A structural fork (*Weichenstellung*) exists only where another defensible decision removes or adds steps; each alternative path (*Lösungsweg*) then has its own steps, its defensibility requirements (*Vertretbarkeitsanforderungen*), and the same BE as the steps it replaces.
- *Auxiliary opinions (Hilfsgutachten).* A step can be marked as owed even when the candidate’s own decision cuts it off, for instance when a contractual claim already fails at formation.

The model states weights, never points.

4.2 Points, checks and grades by code

Code turns the weights into 100 BE in half points, using the largest-remainder method known from seat allocation (Hamilton, in Germany Hare-Niemeyer) [3]: each step first gets the whole half points its weight earns, and the leftover half points go to the steps with the largest remainders. Every step gets at least 0.5 BE, and a heavier step never gets fewer points than a lighter one. A requirement is worth its share of the step, and an alternative path is worth the steps it replaces.

A program, the *validator*, then checks the sheet against fixed rules (Appendix H). Hard rules cover the structure, every cross-reference, the rules for requirements and forks, and the cap of 10 BE per step. If one fails, the generator gets its sheet back with the error messages and tries again, up to 3 times, whereas soft rules only log a warning. The rules compare words, so two steps that say the same thing in other words both pass. Every sheet records the hashes of the validator and of the instructions.

At the end, code adds up the credited points, rounds the total once to half points and maps it to a grade with the exam’s key; several runs can be averaged.

4.3 Judging

Code formats the sheet as text for the judge, which receives the case, the reference solution, the sheet and the answer, each clearly marked, and answers in a fixed format. We test two scoring units. In the *requirement* unit the judge marks each requirement as met, partly met or not met, which earns 1, 0.5 or 0 times its BE. In the *step* unit it awards each step up to its BE in half points, as on a classic sheet. Every credit needs a word-for-word quote from the answer, and the judge follows fixed rules:

- *Placement*. Work counts once, for the step whose requirement it meets, even if it stands elsewhere in the same question and work result. Work on another question counts only if the answer refers to it.
- *Follow-on errors*. Requirements with the consistent standard are judged on the answer's own earlier decisions. Other requirements are judged on the correct law, but an error that has already cost points is not charged again.
- *Auxiliary opinions*. Work on questions that the answer's own decision cut off earns credit like work in the main opinion, but it never replaces the main result.
- *Alternatives*. The judge scores the steps of every path and names, with a quote, the path the answer follows. A path counts only if the answer gives the reasons its defensibility requirements demand, and code counts the named path (design A2).

Further rules cover unforeseen paths, contradictions and padding (Appendix I), which also describes the two other designs for alternative paths that E4 compares. The judge is told to ignore instructions inside its inputs, which a fake answer tests, and it writes short feedback for the candidate that never changes points.

4.4 The quote check and its limits

Code checks that every quote really stands in the answer (Appendix G). It tolerates small differences in layout, inflection and typing, but words that change the meaning must match exactly: negations, result words such as *rechtmäßig* and *rechtswidrig* (lawful, unlawful), qualifiers and numbers. A quote also fails if a negation or a condition stands right next to it, and a credit whose quote fails is set to zero.

The check has limits. It tests that a quote exists, not that it proves the requirement, so a real sentence can be quoted for the wrong requirement. Correct work that the judge paraphrases earns nothing, a sentence copied from the case text passes, and the strict rules at the edges also reject some correct quotes. [PENDING: regression replays at the freeze: quotes re-checked on D1 and D2, and how many flip]

5 Data and design

Table 1 gives the data at a glance.

5.1 D1: the BenGER exams

D1 holds the 15 exams of the BenGER corpus [50]. All four graders used the prior study's 10 general criteria, which its LLM judge also used. The blind graders often disagree: their highest and lowest grade for an answer lie 23.9 points apart on average. Section 6.1 compares with the mean of all 4 grades on the 30 answers written by people, as the prior study did, whereas E2 uses the blind grades as the main reference and the creator's grade as a second one. The prior study's judge was 10.8 points (2.7 grade points) from the mean human grade on average and ranked the answers much as the humans did ($r = .76$). D1 is our only data across many exams, but its reference is noisy.

Table 1. Data at a glance. A second grader grades 20 D2 scripts blind, 10 from each part, and the author grades 5 scripts a second time without seeing the first grade.

	What	Human grades	Role
D1	15 exams (civil, public and criminal law), each with an answer by a person, one by a person with AI help and one by a model	3 blind graders and the exam’s creator, out of 100, on the prior study’s 10 general criteria	Study 1 (Section 6.1)
D2a	15 scripts of one police law exam (LMU), 2024–2025	one grader per script, step by step on the author’s sheet (grader of 9 to be confirmed)	Study 2, development data and first results (Section 6.2)
D2b	23 scripts of the same exam, 2026	the author, blind [PENDING:]	Study 2, held out for one registered test
Z2	Gregor Roth’s sheet and grading workbook for a civil-law exam	none	Study 2, sheet comparison (E5)

5.2 D2: the expert exam

D2 is one police-law exam from LMU, set each year from 2024 to 2026, whose author is a co-author of this paper. The author’s sheet has 46 steps worth 100 BE in half points, and the author’s key passes a script at 40%. Some step weights changed between years, so the sheet exists in 3 versions with the same steps; the judge always sees the current one.

- *D2a* holds 15 typed scripts, 3 from 2024 and 12 from 2025, graded step by step on the sheet version then in use. We call these *reference grades*, because for 9 scripts the grader is still being confirmed.
- *D2b* holds 23 student scripts from 2026, written on our platform as a practice exam; we drop 2 uploads from the author’s own account (a test upload and a copy of a *D2a* script). The platform graded the scripts on submission with GPT-5.4 Mini, and the students saw these grades. The author grades them blind, without seeing any model grade. [PENDING: blind expert grades for D2b]

D2 thus has 38 distinct scripts, or 37 if one pair that shares text counts once. We lack the 2024 case text, so the 3 scripts of 2024 are judged against a later one. [PENDING: the 2024 case text from the exam author]

All D2 texts were pseudonymized and checked for personal data before any research call to a model provider (Appendix J). No generator and no judge without a sheet sees the author’s sheet, and a canary check confirms that the fixed instructions and examples in every prompt contain no terms from the author’s sheet or exam.

Z2 is Gregor Roth’s sheet and grading workbook for a civil-law exam, so far without case, reference solution or scripts. E5 generates sheets for this exam and compares them with Roth’s sheet. [PENDING: case and reference solution from Roth]

5.3 Experiments

D1 and D2a are development data. D2b is held out: it holds new scripts of the same exam, not a new exam, which the author grades from exported files before seeing any generated sheet or model grade, and one registered test runs on them once. [PENDING: the registered hypothesis and its success criterion] Stability needs no human grade, so E1 uses all 38 D2 scripts. Table 2 lists the experiments. The baselines are grading without a sheet, on the prior study’s 10 general criteria, and grading on the exam author’s own sheet. Since the author’s sheet has no requirements, the

Table 2. Experiments of Study 2. [PENDING: actual costs per experiment]

Experiment	RQ	What we do
E0 Freeze and gate	RQ1, RQ2, RQ3, RQ4	Freeze one version of the pipeline. Test every judge and sheet with fake answers in 3 runs. Register the D2b test before any D2b grade exists.
E1 Steadier?	RQ1	Grade all 38 D2 scripts with no sheet, the author’s sheet and 10 generated sheets (2 models, 5 each, scored per requirement and per step): Luna 3 runs, DeepSeek-V4-Pro and GPT-5.4 Mini on part of them. Repeat on D1 (45 answers, 3 runs). Split the spread into its sources, count pass/fail flips between runs, compare with averaging runs at equal cost.
E2 Close to the examiner?	RQ2	Compare the E1 grades with the author (D2a, and D2b as the registered test), a second grader (20 scripts, incl. the 9 with an unconfirmed grader) and the author’s second grading of 5 scripts. On D1, compare with the blind graders and test whether a model could replace one of them. Count wrong passes and fails at pass marks from 35 to 55%.
E3 Which models?	RQ3	5 generators with 3 sheets each on D2 and 5 D1 exams, one judge. 4 judges on fixed sheets, one of them an open model and one never used in development. Cost per valid sheet. The author reviews sheets without knowing which model wrote them.
E4 Doctrine and safety	RQ4	Fake answers the author writes, with the expected loss fixed in advance: a defensible other path, a consistent follow-on error, reversed decisive results, fluent but wrong application, padding, paraphrase, an answer written by a model, cardinal errors. Three designs for alternative paths. 200 quotes checked by hand.
E5 Second exam	RQ3, RQ4	Sheets generated for Roth’s exam, compared with Roth’s sheet (coverage, weights) and checked by Roth. Needs Roth’s case and reference solution.

judge scores it per step, whereas generated sheets are scored both ways so that the sheet and the scoring unit can be told apart. The whole program costs USD 140 to 200 at list prices (Appendix J).

Measures. Accuracy is the average distance of a total from the human grade (mean absolute error, MAE) and its sign (bias: too generous or too strict). We also give the error in grade points on the 0 to 18 scale and the number of students a model passes or fails differently from the human grade. For stability we split the spread of totals into real differences between scripts and noise from the judge, from the sheet a generator happens to write and from repeated runs [5]. Dependability, from 0 to 1, is the share of the spread that reflects real differences between scripts when a single run decides. We also count how often a second run lands on the other side of the pass mark [44] and compare sheets with averaging runs at equal cost. Each human reference was graded with one method, D1 on the 10 general criteria and D2 on the author’s sheet, so each favours the grading basis it shares.

Fake answers (probes). Before any benchmark run, every judge and sheet must clear a set of fake answers in each of 3 runs (Appendix F): empty, copied, off-topic or manipulative answers must score about zero, and the reference solution must score high. Cutting one section of the reference solution must cost about its BE, while moving correct reasoning under another heading should cost almost nothing.

Uncertainty. Apart from the registered test, we estimate rather than test. Every range is a 95% interval from redrawing exams (D1) or scripts (D2) many times (bootstrap), and with 15 graded scripts only large differences are visible (Appendix E). Each judge runs at one fixed temperature in all comparisons, and scripts that the judge flags for review keep their totals in our analyses.

6 Results

Totals are points out of 100, and a grade point is one step on the 0 to 18 scale. On the expert exam, 4 points make one grade point from the pass mark (40 points) up and 10 below it. r says whether two graders rank the scripts in the same order (1 = the same order) and ignores a constant shift. Brackets give 95% intervals; if the interval of a difference includes zero, we cannot tell whether the difference is real.

6.1 Study 1: a step-score sheet on 15 exams

Study 1 tested a step-score sheet on D1, in which each step carried a weight class, ordered levels of partial credit and follow-on-error rules. 12 generators wrote one sheet per exam, and a random draw picked one sheet per exam for all judges. 7 judges graded the 45 answers 3 times with that sheet (2 more once), and 6 of them also graded without a sheet, on the prior study's 10 general criteria [50]. Totals are sums of the step scores (Appendix C).

Writing sheets worked. 170 of 180 pairs of generator and exam yielded a valid sheet, and 10 of the 12 generators covered every exam, but the sheets differed widely: the mean number of steps per generator ranged from 6 to 26 (Table 10).

The sheets did not make grading steadier (Table 9). Graded again, Luna's total for the same answer moved by 2.3 points without a sheet and 2.0 with one (standard deviation over runs). Across the 5 judges other than GPT-5.4 Mini, the sheet changed this spread by +0.02 points on average $[-0.63, +0.73]$, and Claude Sonnet 4.6 was steadier without a sheet (Table 12); with GPT-5.4 Mini included, the average would be +0.29. GPT-5.4 Mini's own spread grew from 3.9 to 5.5 points, although its interval includes no change. The judges disagreed with one another far more than with themselves: the standard deviation of the 6 judges' totals for the same answer was 7 to 10 points.

Nor did the sheets bring the grades closer to the humans. With the sheet, Luna ranked the answers about as before ($r = .77$ against $.80$) but turned from 4 points too generous to 7 points too strict, and GPT-5.4 Mini's average error rose from 11 to 19 points.

The sheets invited credit for nothing. With a sheet, Luna gave a near-empty answer about 96 of 100 points in 26 of 45 runs and never did so without one, while GPT-5.4 Mini gave empty answers a median of 95 points. Only Claude Opus 4.7 and DeepSeek-V4-Pro kept every fake answer near zero on every run (Table 14).

We attribute these failures mainly to the Study 1 pipeline, which checked no quotes, listed the steps out of outline order and gave the steps of alternative paths no fields of their own (Appendix C.1). The pipeline of Section 4 checks every quote, keeps the outline order and gives every path its own fields, and E0 and E3 test whether weaker judges can then use a sheet safely.

6.2 Study 2: first results on the expert exam

These preliminary runs are not the benchmark (E1 to E5). They cover the 15 D2a scripts, sheets from two generators and up to three judges, and a version of the output specification with fewer validator checks than Section 4 describes. With 15 graded scripts, only differences in average error of about 2.5 to 8.4 points can be told apart from chance (80% power, Appendix E), so these are first results, not a ranking. The yardstick is the reference grades of the D2a scripts, which were given on the examiner's own sheet and so favour that sheet over generated ones; for 9 scripts the grader is still being confirmed. The reference fails 5 of the 15 scripts.

Agreement with the reference grades (Table 3). Without a sheet, Luna was off by 14.4 points on average (3.3 grade points), almost always too generous (+12.4), and it passed 5 of the 5 students whom the reference fails. On the examiner's sheet the same judge was off by 5.1 points (1.1 grade points) and ranked the scripts almost as the reference did ($r = .95$); against grading without a sheet,

Table 3. Expert exam: judge totals against the reference grades of the 15 D2a scripts, in points out of 100 (from the pass mark of 40 points up, 4 points make one grade point). Generated sheets are scored per requirement; Table 16 adds per-step scoring. Each value is computed per run and averaged over the runs. Grade points: the average error on the 0 to 18 scale under the exam’s key. Wrongly passed: students the judge passes whom the reference fails, and wrongly failed the reverse. Run to run: the average difference between two runs on the same script. [†] not checked with the fake answers. [‡] failed that check (Table 18).

Grading basis	Judge	Runs	Average error [95%]	Grade points	Generous (+), strict (−)	<i>r</i>	Wrongly passed / failed	Run to run
No sheet	Luna [‡]	3	14.4 [10.2, 18.5]	3.3	+12.4	.84	5 / 0	1.8
No sheet	Mini [†]	1	13.1 [8.8, 17.8]	3.0	−2.8	.39	5 / 2	−
Examiner’s sheet (per step)	Luna	3	5.1 [3.3, 7.0]	1.1	−3.4	.95	1 / 0.3	2.4
Examiner’s sheet (per step)	DeepSeek	1	5.2 [3.2, 7.4]	0.9	−3.9	.95	0 / 1	−
Examiner’s sheet (per step)	Mini [†]	1	9.4 [6.2, 13.0]	1.7	−1.6	.74	3 / 2	−
GPT-5.4 sheet	Luna	3	6.1 [4.2, 8.2]	1.3	+2.2	.91	2.3 / 0	4.1
GPT-5.4 sheet	DeepSeek	1	6.2 [4.1, 8.4]	1.4	+3.1	.91	3 / 0	−
GPT-5.4 sheet	Mini [†]	1	10.3 [6.6, 14.4]	2.3	+5.7	.74	4 / 1	−
GPT-5.4 Mini sheet 1	Luna	3	7.3 [4.9, 10.1]	1.6	+6.1	.92	3 / 0	3.2
GPT-5.4 Mini sheet 2	Luna [†]	3	7.3 [4.8, 10.0]	1.7	+7.2	.94	4 / 0	3.2

Table 4. Expert exam, stability of GPT-5.6 Luna on all D2 scripts (no human grades needed), in points out of 100. Run to run: the standard deviation of a script’s total between runs. Between scripts: the spread of the scripts’ true totals. Dependability: the share of the spread that reflects the scripts when one run decides (0 to 1). Flips: how often a second run lands on the other side of the pass mark. A new sheet each time: one run with a freshly generated sheet, so the sheet adds its own noise, and flips there compare two sheets. Settings run on fewer scripts: Table 17.

Grading basis	Scripts	Run to run	Between scripts	Dependability	Flips	US cents per run
No sheet	38	2.9	9.6	.91	0%	0.6
Examiner’s sheet, per step	38	3.4	16.2	.96	9%	1.2
GPT-5.4 sheet, per requirement	38	2.9	15.2	.97	4%	1.6
GPT-5.4 sheet, per step	38	3.6	15.2	.95	5%	1.4
GPT-5.4 Mini sheet 1, per step	38	3.0	13.9	.96	2%	1.0
GPT-5.4 Mini sheet 2, per step	38	2.7	15.2	.97	4%	1.1
A new GPT-5.4 Mini sheet each time, per step	38	4.0	14.3	.93	6%	−

its error changed by −9.3 [−14.5, −3.9] points. With generated sheets it was off by 5.7 to 8.3 points, mostly because they were more generous, so every pass/fail disagreement was a wrong pass. The GPT-5.4 sheet, scored per step like the examiner’s, came as close as the examiner’s own sheet (difference +0.6 [−2.3, +3.7]), and scoring per requirement instead of per step changed the error by at most 0.9 points (Table 16). The platform’s default judge, GPT-5.4 Mini, was 9.4 points off even on the examiner’s sheet and passed or failed 5 of 15 students differently; without a sheet it ranked the scripts poorly ($r = .39$) and passed or failed 7 of them differently.

Stability. The sheets did not make repeated runs steadier (Table 4): without a sheet, Luna’s total moved between runs by 2.9 points (standard deviation), on the examiner’s sheet by 3.4 and on generated sheets by 2.6 to 3.6. On D2a, the examiner’s sheet changed the run-to-run spread by +0.6 [−0.5, +1.9] points against no sheet. The steadiness without a sheet came with a compressed scale, however. Luna spread the D2a scripts with a standard deviation of 8.0 points against the reference’s 17.5 and passed every D2a script, so a second run could not flip anyone: a grader can be steady and still wrong. A new sheet from the same generator moved a script’s total more than a new run did. One run with a fresh GPT-5.4 Mini sheet had a standard deviation of 4.0 to 4.1 points,

and 6% to 16% of the scripts passed on one sheet and failed on the other (Table 17). Averaging three runs without a sheet would lower the run-to-run spread to 1.7 points at about 1.9 US cents, but it cannot fix the generous level.

Fake answers. With every sheet, every judge we checked kept all six fake answers at zero (Table 18 in Appendix D), whereas without a sheet Luna gave them up to 14 points and failed the check, since the judge without a sheet has no quote check. The rows of Table 3 that were not checked, or failed, are marked. Removing one part of the answer key cost about as many points as the sheet gives that part, and moving a correct paragraph under another heading cost almost nothing. On the examiner's sheet, scored per step, Luna still proposed half a point for every step of an empty or off-topic answer, and only the rule that a credit needs a quote removed these points. Reversing the results of the answer key cost only 5 to 18 points with a sheet but 38 without one, because the requirements credit the reasoning whatever the result. What such an answer should lose depends on how an examiner grades a conclusion that contradicts its own reasoning, which E4 fixes with the examiner in advance.

6.3 Study 2: experiments E1 to E5

[PENDING: results of E1 to E5, each error given in grade points and as students wrongly passed or failed]

7 Discussion

[DRAFT: the answers per research question are written after E1 to E5]

What Study 1 showed. A generated sheet did not by itself make model grades steadier. With a detailed sheet, several judges also gave nearly full marks to answers that were empty or only repeated the facts of the case, a failure that an average agreement figure hides. Every judge must therefore pass tests with fake answers before use, and code must check that each point rests on words that stand in the answer.

What the expert exam suggests so far. On the examiner's own sheet, the best judge was about 5 of 100 points from the reference grades on average, or 1 grade point, and ranked the scripts almost as the reference did. Without a sheet the same judge was steady but about 12 points too generous: it passed every student whom the reference fails and gave fake answers up to 14 points. Generated sheets came close to the examiner's sheet (6 to 8 points off), though mostly more generous. On this exam, then, a sheet did not make the judge steadier, but it made its grades far more accurate. The platform's default judge was about 9 points off even on the examiner's sheet. Because a new sheet from the same generator moved a script's total more than a new run did, a sheet should be fixed and approved once per exam. These are first readings on the scripts of one exam, and they favour the examiner's sheet (Section 8).

Answers to the research questions. [PENDING: one paragraph per research question after E1 to E5: first a plain answer (yes, no, partly, not yet), then what it means in grade points and in students wrongly passed or failed, then the number and its interval]

What this means for exam practice. Three points follow, all provisional. *Equal opportunity.* A newly generated sheet can move a script's total by several points, so a sheet should be generated once per exam, checked and approved by the examiner, and then used unchanged for every candidate. *Accuracy.* The generated sheets tended to be generous, so in practice feedback a student may believe they would pass when they would not; such feedback should say so, or the sheet should first be calibrated on a few scripts the examiner has graded. *Reasons.* Each point comes with a quote that code has found in the answer, so an examiner or a student can see where it was earned. The code confirms only that the words are there, not that they meet the requirement, so the quotes can support the examiner's reasons but cannot replace the examiner's own evaluation, which the

examiner must make personally [8]. The evidence so far supports feedback on practice exams with these safeguards. A second reader that points the examiner to scripts where the totals differ is plausible but untested, and the evidence does not support grading exams that count towards a degree without an examiner. EU law classes AI that evaluates learning outcomes as high-risk (AI Act, Annex III, point 3(b)), with the duties applying from December 2027 [19, 20]. Grading one script once costs about 1 to 17 US cents with the judges we priced (Appendix J). **[PENDING: recommended pipeline and its cost per script]**

8 Limitations

One expert exam. D2 is one case in one area of law, graded on its author’s sheet, and D1 spans many exams but has a noisy human reference. Results on D2 may not transfer to other exams, areas or examiners, and the second grader gives a human ceiling for one exam only.

Each yardstick favours one grading basis. The D2a reference grades were given on the examiner’s own sheet, and one setting grades with that same sheet, which favours it over generated sheets. On D1 all human graders used the 10 general criteria that the judge without a sheet also uses, which favours grading without a sheet. The second grader and the registered D2b test reduce this bias but do not remove it.

Open data questions. For 9 D2a scripts the grader is not yet confirmed, the 3 scripts of 2024 are judged against a later year’s case text, and the judge sees the current sheet also for scripts graded on an older version of it. The D2b students saw model grades from the platform; the author grades their scripts without seeing them.

The exam author is a co-author. The author wrote the exam, grades the scripts, reviews generated sheets and co-writes this paper, and so knows the study’s aims. Blind grading of D2b and the second grader will limit this; the author also reviews generated sheets without knowing their source (E3), but this review is not independent.

Development on the test exam. The pipeline was developed on D2a while the author’s sheet was available. The prompts contain no examples from D2, and code checks every prompt for words from the author’s sheet, but the check matches words only and cannot rule out that the sheet shaped the prompts. Results on D2a are therefore optimistic.

Models and settings. GPT-5.6 Luna cannot be set to repeat its output, and even at its least random setting DeepSeek-V4-Pro’s totals varied between runs (run-to-run spread 1.81 in Study 1), so the spread mixes the model, its settings and the provider’s servers. Providers change and retire models; we record model ids and dates.

Exploratory design. Only the test on D2b is registered in advance, and it runs once; everything else is exploratory. We report many comparisons without correcting for their number, and Study 1 added several settings after seeing earlier results.

Scope. All D2 scripts were typed, and handwritten exams would need transcription, which we did not test. Nor did we test how examiners use a generated sheet or a model grade, for instance whether they correct it or simply accept it.

9 Ethics

Consent and personal data. The students whose D2 scripts we use consented to research use, for the 15 scripts of D2a and the 25 platform submissions of D2b. The consent does not extend to sending their personal data to model providers. All D2 texts are therefore pseudonymized and checked for personal data before any research call (Section 5.2). The platform grades that the D2b students saw were produced in the normal use of the platform, before this study. **[PENDING: owner to confirm the wording on the platform-side grading and the providers’ data terms]** D1 reuses the prior study’s corpus under the consent obtained there, with pseudonymized graders [50].

Students who saw LLM grades. The D2b students received LLM grades as feedback on a practice exam. [PENDING: owner: whether and how students learn the expert grades]

Roth's sheet. Gregor Roth agreed to be named and to have his sheet and workbook used. We report only their structure, not their content.

10 Data and code availability

The platform is open source under the Apache 2.0 license at <https://github.com/SebastianNaglbenger-platform> [PENDING: tagged release or commit, archived with a DOI]. It contains the judge, the quote check and the aggregation, whereas the generator's instructions and output specification, the validator, the allocation of points and the rendering of generated sheets belong to a proprietary extension of the platform. [PENDING: owner decision: release of the generator instructions, output specification and validator rules] The folder publications/Transformation holds the analysis scripts and every aggregate this paper reports, and every run records the hashes of its code, prompts and validator. Recomputing the aggregates needs data we do not release: the per-call records of D1, which contain third-party exam text, and the private D2 and Z2 material. D1's exam texts and the generated sheets, which restate the reference solutions, are not redistributed, and for D2 and Z2 we publish only aggregates (Appendix J).

11 AI usage

This section covers AI tools used to do the research and write the paper, not the models under study. For Study 1, the authors used OpenAI GPT-5.6 Sol to discuss approaches and Anthropic Claude Fable 5 to help write code and text. For Study 2, Anthropic Claude Opus 5.5 worked as a coding agent in the authors' repositories under the first author's direction. It wrote the pipeline of Section 4 (output specification and validator, checklist judge, quote check), the experiment runner and all analysis scripts, and it ran the experiments within budgets the first author approved one by one. It also drafted most of the text, tables and appendices, including this section, and it searched the literature and checked the references it added against their sources. Further instances of the same model, given identical instructions, reviewed the code, the data handling, the statistics and the text in 4 rounds, simulated a peer review, and reviewed each section for readability and missing literature. Because the agent ran the scripts that extracted and pseudonymized the D2 texts on the authors' machine, parts of the raw scripts may have reached Anthropic's service before pseudonymization. [PENDING: authors: confirm, and name the terms under which this happened] [PENDING: authors: who wrote Section 2 and the legal parts of the discussion] The authors checked [PENDING: every reference against its source; every number in the abstract, discussion and tables; the quote check and the allocation code]. The authors set the research questions and the design, supplied the exam, its grades and the sheets, decided every change of scope and spending, and take full responsibility for the paper.

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Table 5. German terms used in this paper and their glosses.

Term	Gloss
<i>Falllösung</i>	Written case solution: the end-to-end work product that is graded.
<i>Sachverhalt</i>	Fact pattern: the facts of the case the candidate works on.
<i>Gutachtenstil</i>	Appraisal style: the conclusion follows the reasoning; the register graded here.
<i>Mustertlösung</i>	Expert reference solution against which submissions are graded.
<i>ausreichend</i>	“Sufficient”: lowest passing grade (4-6 of 18 points).
<i>Bewertungsbogen</i>	Grading sheet: the expected solution split into steps, each worth a number of BE.
<i>Bewertungseinheiten (BE)</i>	Assessment units: points on a grading sheet, 100 per exam here, in half points.
<i>Notenpunkte</i>	Grade points: the statutory scale from 0 to 18 of German law examinations.
<i>Notenstufen</i>	The seven named grades of that scale.
<i>Staatsexamen</i>	State examination: the examination that gives access to the legal professions.
<i>Chancengleichheit</i>	Equal opportunity: comparable candidates must be graded by comparable standards.
<i>Klausurschwerpunkte</i>	The exam’s focal problems.
<i>Maßstab</i>	The standard a requirement is judged by: <i>richtig</i> or <i>folgerichtig</i> .
<i>richtig</i>	Correct: judged against the correct law.
<i>folgerichtig</i>	Consistent: judged against the candidate’s own earlier decision.
<i>vertretbare Lösung</i>	Defensible solution: one argued with sound reasons, which must not be marked as wrong.
<i>Folgefehler</i>	Follow-on error: an error that later steps build on; it is not penalised twice.
<i>Weichenstellung</i>	Structural fork: a defensible decision that changes the further structure of the solution.
<i>Lösungsweg</i>	Alternative path after a structural fork, with its own steps.
<i>Vertretbarkeitsanforderungen</i>	The reasoning that makes an alternative path defensible.
<i>Hilfsgutachten</i>	Auxiliary opinion: the part a candidate still owes after a decision that ends the case.
<i>Arbeitsergebnis</i>	Work result, such as an opinion or a draft judgment.
<i>Kardinalfehler</i>	Cardinal error: an error grave enough to limit the grade as a whole.

A Glossary

Table 5 glosses the German terms and Table 6 the technical terms.

Table 6. Technical terms used in this paper.

Term	Meaning
Judge	A language model that grades an answer.
Generator	A language model that writes a grading sheet from the case and its reference solution.
Code	A fixed program rather than a model, which computes points, checks rules and quotes, and adds up totals.
Run	One grading of one answer; the same judge can grade the same answer several times.
Run-to-run spread	How far a judge’s total for the same answer moves between runs (standard deviation).
Average error (MAE)	The mean distance between a model’s total and the human grade, in points.
Bias	The mean signed distance, positive if the model is too generous and negative if too strict.
r	Whether two graders rank the answers in the same order (1: the same order), regardless of a constant shift.
κ	Agreement on pass or fail, corrected for the agreement expected by chance.
95% interval	The range in which the true value lies with 95% confidence, from redrawing the exams or scripts many times.
Dependability	The share of the spread of totals that reflects real differences between scripts (0 to 1).
Fake answers (probes)	Made-up answers, such as an empty one or a copy of the case, that must score about zero.
Quote check	Code that tests whether each quote of the judge stands in the answer.
Validator	Code that checks a generated sheet against fixed rules and sends errors back to the generator.
Scoring unit	What the judge scores, either each requirement (met, partly met, not met) or each step (points).

B Grading paradigms compared

Table 7 sets the three grading paradigms of Section 2 side by side with our sheet.

Table 7. Three grading paradigms and our sheet. D2 is the exam author’s sheet and Z2 Gregor Roth’s sheet and workbook (so far without case or scripts). Partly fixed is a common practice that leaves a share of the BE to the grader (typical values, – = not set). ? = to be completed with the examiners. [PENDING: rows on alternatives and result items, confirmed with both examiners]

	Exam author (D2)	Roth (Z2)	Partly fixed	Our sheet
What is counted	absolute BE	% of the item above	BE per step, plus a free share	BE, allocated by code
Total	100 BE	100%	100 BE, 80% fixed	100 BE
Smallest items	46 steps	40 items, up to 4 levels deep	steps for the fixed share	steps with 1–6 requirements
How an item is scored	BE in half points	grade 0–18, whole points	BE per step, the free share at discretion	per requirement (met, partly, not) or per step
Largest step	10 BE	9% of the grade	–	cap 10 BE
From items to total	sum of BE	weighted average of item grades	sum of BE	sum of BE
Pass mark	40% of BE	grade 4	the exam’s key	the exam’s key
Adjustment	?	manual bonus or malus	the free 20%	none
Credit for bare results	none (some titles state results)	1 of 4 weighted	–	only with its reasoning
Graders per script	one (see Section 5)	? (2 final-grade fields)	–	one LLM judge, run k times
Alternatives	?	?	–	requirements or forks

C Study 1 details

This appendix gives the details of Study 1 that Section 6.1 summarizes: the properties of its pipeline that affect its numbers, its results per setting, generator, judge and fake answer, and an audit of how its sheets allocated points.

C.1 Properties of the Study 1 pipeline

The Study 1 pipeline differs from the pipeline of Section 4 in the ways listed in Table 8. Where a property touches a Study 1 number, this paper reports the number without it, for instance totals as sums of the step scores.

C.2 Results

Generation (RQ3). 170 of 180 pairs of generator and exam hold a valid sheet, and 10 generators covered every exam. Llama-4 Maverick covered 12 exams and GPT-5.4 Mini 8, mostly because feedback did not repair broken references of alternative paths. Of 220 dispatched series, 50 failed: 34 aborted at the provider (all GPT-5.4 Mini, credit errors), 14 failed validation and 2 failed parsing. First-attempt compliance over all series that reached the model spans 13% to 100%. Counted over valid sheets only, GPT-5.4 Mini looks better than it is, with 38% there against 17% over the series that reached the model (3 of 18) and 6% over all 52 of its series (Table 10). The generators did not converge: mean step counts range from 6.3 to 25.5, and stated weight sums from 34 to 179. A lexical screen found vague formulas that the prompt itself bans in 7 of 170 sheets (16 with near variants).

Variance (RQ1). For Luna, the run-to-run SD falls from 2.25 without a sheet to 1.96 with the zero-shot sheet and 1.78 with a sheet written from an example (few-shot). The direction is consistent, but no single step is established: few-shot minus no sheet is -0.47 $([-1.02, +0.07]$, sign test $p = .37$), zero-shot minus no sheet -0.29 $([-0.77, +0.15]$, $p = .54$), and few-shot minus zero-shot -0.18 $([-0.65, +0.24]$, $p = 1.00$). Across the 5 judges other than GPT-5.4 Mini that graded both ways, the zero-shot

Table 8. Properties of the Study 1 pipeline that affect its numbers.

Topic	Property
Allocation	The routine broke the quota rule for 27 steps in 11 sheets, with 12 order inversions in 3. Of the 15 drawn sheets, 1 is affected (DeepSeek-V4-Pro, where 8 of its 24 steps change by one point), and all 11 affected sheets enter the generator ranking (Appendix C.3).
Totals	The stored total was sometimes the model’s own, within half a point of the step sum. With step sums the zero-shot run-to-run SD moves for Luna from 1.97 to 1.96, for Mini from 5.52 to 5.51, and few-shot for Luna from 1.80 to 1.78.
Judge schema	Response keys followed storage order (by key length), not outline order.
Alternatives	Steps of alternative paths had no response keys, so a judge had to put credit for another path onto the replaced steps, where the per-step cap dropped any overflow. [PENDING: count of affected paths]
Quote check	No quotes were checked, and the system prompt was generic.
Prompt wording	The generator prompt asked for exactly 100 points and also for relative weights, and the weight class was shown to the judge but played no part in the arithmetic.
Few-shot pool	The few-shot setting changed the generator pool along with the exemplar, so the two effects are confounded.

Table 9. Study 1 (D1): two judges with and without a generated sheet, in points out of 100 (about 3 points make one grade point above the pass mark). Run-to-run spread: standard deviation of a judge’s total over 3 runs on the same answer (45 answers). The other columns compare the first run with the mean of the four human grades (three blind graders and the exam’s creator) on the 30 answers written by people.

Judge, grading basis	Run-to-run spread	Same order (<i>r</i>)	Average error	Too generous (+) / strict (–)
Luna, no sheet	2.3	.80	11.2	+4.3
Luna, generated sheet	2.0	.77	12.8	–6.9
Luna, sheet from an example	1.8	.70	15.8	–7.9
GPT-5.4 Mini, no sheet	3.9	.75	11.2	–2.7
GPT-5.4 Mini, generated sheet	5.5	.41	19.4	+7.4
GPT-5.4 Mini, sheet from an example	5.2	.36	21.3	+5.8

effect averages +0.02 [–0.63, +0.73], Table 12), an unweighted mean of the per-judge differences without correction for multiple comparisons. Only Sonnet’s per-judge interval excludes zero, and it points the adverse way: Claude Sonnet 4.6 is steadier without a sheet (+0.34, [+0.08, +0.64]). For the weakest judge, GPT-5.4 Mini, the run-to-run SD rises from 3.87 to 5.51 (+1.64, [–0.18, +3.45], clustered by exam, cell-level sign test $p = .02$, Wilcoxon $p = .007$). The steadiest few-shot sheets, a post-hoc subset by generator, reached 1.26 with Luna, and an in-sample floor (the best sheet per exam, chosen after the fact) 1.01; both are optimistic bounds without intervals. These contrasts use step sums, and on the stored totals they differ by at most 0.02.

Agreement (RQ2). Against the mean human grade, Luna’s r moved from .80 to .77 and .70, and its bias flipped from +4.3 to –6.9 and –7.9; the few-shot exemplars had been chosen for stability alone. Its zero-shot MAE rose by +1.6 [–3.3, +7.0]). GPT-5.4 Mini graded adequately without a sheet ($r = .75$, MAE 11.2) but not with the sheet ($r = .41$, MAE 19.4, an increase of +8.2 [+2.0, +15.8]), and few-shot sheets left it there ($r = .36$, MAE 21.3). Leave-one-exam-out bias correction shows that the error of the prior study’s control judge was mostly a level shift (MAE 14.9 to 9.5), whereas Luna’s zero-shot error was not (12.8 to 13.1).

Judges and fake answers. The 7 roster judges graded all answers against the drawn sheets (Table 11). Apart from Mini they reached $r .70$ to $.77$, and Claude Opus 5 and GPT-5.6 Terra, one run each,

Table 10. Study 1, generation by generator: valid sheets, first-attempt compliance over successful sheets and over all series that reached the model, mean attempts per valid sheet, and mean step count.

Generator	Valid	1st (valid)	1st (series)	Attempts	Steps
Claude Opus 4.7	15/15	73%	73%	1.27	17.1
Claude Sonnet 4.6	15/15	13%	13%	1.93	22.0
DeepSeek-V4-Flash	15/15	80%	71%	1.27	18.5
DeepSeek-V4-Pro	15/15	100%	100%	1.00	25.5
GPT-5.4	15/15	100%	100%	1.00	16.8
Gemini 3.1 Flash-Lite	15/15	80%	80%	1.20	6.3
Gemini 3.1 Pro	15/15	100%	100%	1.00	15.3
Qwen3-235B-Thinking	15/15	40%	40%	1.60	12.8
Qwen3.5-122B	15/15	13%	13%	1.93	9.3
Qwen3.6-35B	15/15	60%	56%	1.60	11.5
Llama-4 Maverick	12/15	75%	60%	1.25	6.5
GPT-5.4 Mini	8/15	38%	17%	1.88	17.1

reached .74 and .75. A set of 45 fake answers per type (the case text verbatim, a near-empty answer, another exam’s reference solution) and each exam’s own reference solution as a positive control tested execution (Table 14). Only Claude Opus 4.7 and DeepSeek-V4-Pro cleared every negative fake answer in every run. Luna credited the near-empty answer in 26 of 45 zero-shot runs and 24 of 45 few-shot runs, with 96.2 and 97.1 points on average when it did (median 100.0 and 99.5), but in 0 runs without a sheet. GPT-5.4 Mini failed 100 of 135 runs on negative fake answers with the sheet and 125 without one, but its failures without a sheet were small (median 5 points on the empty answer) whereas its failures with the sheet were near-total (median 95). With the sheet, the off-topic answer scored at most 4.5 points with every judge except Mini (up to 94). All 7 judges passed the positive control with the sheet, while Mini failed it without a sheet in 13 of 45 runs. On one answer that only restates the case, Mini with the sheet gave 99, 100, 100 points and Luna 0, 0, 0.

Generators (RQ3). A crossed sweep graded each exam’s answers under every valid sheet of that exam, 170 sheets and 510 combinations of sheet and answer, with Luna over 3 runs (Table 13). DeepSeek-V4-Flash sheets lowered agreement the most ($r = .40$), and Llama-4 Maverick sheets made Luna least stable (run-to-run SD 4.07). GPT-5.4 Mini, the weakest judge and least compliant generator, ranked among the best sources on the exams where it produced a sheet (paired Δ MAE -3.2 on 8 exams), but its probability of ranking first is .47 and the estimate is conditional on producing a sheet. The ranking replicates moderately with Claude Sonnet 4.6 as judge (one run, Spearman .50, and .86 among full-coverage generators). Few-shot generation raised Mini’s valid sheets from 8 to 10 of 15, but since the few-shot setting also used its own pool of 5 generators, its effects are confounded with that pool.

C.3 Allocation audit

The audit re-derives the points of all 170 Study 1 sheets (2542 steps) from their stated weights under each routine of Table 15. A step lies outside its fair range when its points differ from its quota by a full unit or more, and an order inversion is a pair in which the lighter step receives more points. The Study 1 routine is reproduced exactly by a copy of its code. The routine of Section 4.2 sets steps whose share falls below the minimum to it before the remainders are ranked and apports the rest again, so every other step lands within one half point of its exact share, and weights that already sum to the total pass unchanged.

Table 11. Study 1, judges on the drawn sheets (step sums), sorted by MAE. r and MAE: first run against the mean human grade (30 answers). SD_{rep} : run-to-run SD over 3 runs on the listed number of answers. Battery: fail(N) counts failing runs on negative fake answers. P01: the first run on an answer that only restates the case. Below the line: judges with one run, from the database extraction.

Judge	r	MAE	SD_{rep}	Answers	Battery	Pos.	P01
GPT-5.6 Luna	.77	12.8	1.96	45	fail(32)	pass	0
Qwen3.5-397B	.74	13.2	5.43	45	fail(19)	pass	0
Gemini 3.1 Pro	.73	13.6	2.12	44	fail(20)	pass	0
Claude Sonnet 4.6	.77	15.2	1.17	45	fail(6)	pass	0
Claude Opus 4.7	.74	15.5	1.25	36	pass	pass	0
GPT-5.4 Mini	.41	19.4	5.51	45	fail(100)	pass	99
DeepSeek-V4-Pro	.70	19.7	1.81	45	pass	pass	0
Claude Opus 5	.74	14.1	–	–	–	–	0
GPT-5.6 Terra	.75	14.4	–	–	–	–	0

Table 12. Study 1, run-to-run SD of each judge with the zero-shot sheet and without a sheet on the same 45 answers, 3 runs each (step sums). 95% intervals from a bootstrap that redraws exams, and a sign test on answers. The pooled estimate averages the judges other than GPT-5.4 Mini without weights and without correction for multiple comparisons.

Judge	Sheet	No sheet	Δ	95% interval	Sign p	Pooled
GPT-5.6 Luna	1.96	2.25	–0.29	–0.77 to +0.15	.54	yes
Claude Sonnet 4.6	1.17	0.83	+0.34	+0.08 to +0.64	.64	yes
Gemini 3.1 Pro	2.14	2.42	–0.28	–0.83 to +0.28	.88	yes
DeepSeek-V4-Pro	1.81	1.67	+0.13	–0.48 to +0.65	1.00	yes
Qwen3.5-397B	5.43	5.22	+0.21	–3.09 to +3.80	.37	yes
GPT-5.4 Mini	5.51	3.87	+1.64	–0.18 to +3.45	.02	no
Pooled (unweighted)			+0.02	–0.63 to +0.73		

Table 13. Study 1, sheet sources ranked by outcome with Luna as judge: agreement, MAE, signed bias, run-to-run SD, and the within-exam paired Δ MAE (negative: better than the exam’s average source), with the bootstrap median rank and its 95% interval. Right: MAE and bias with Claude Sonnet 4.6 as judge, one run.

Source	Sheets	r	MAE	Bias	SD_{rep}	Δ MAE	Rank	MAE _S	Bias _S
Gemini 3.1 Flash-Lite	15	.82	11.4	+2.0	2.40	–2.4	2 [1–10]	13.7	–5.9
Qwen3-235B-Thinking	15	.81	12.0	–5.5	2.19	–1.9	3 [1–8]	13.2	–11.2
Qwen3.6-35B	15	.78	12.5	–4.9	2.21	–1.4	4 [1–8]	13.8	–10.2
GPT-5.4	15	.80	12.6	–6.2	1.80	–1.2	5 [2–8]	14.7	–10.8
Qwen3.5-122B	15	.76	12.8	+1.3	1.73	–1.1	5 [1–10]	14.1	–5.8
Gemini 3.1 Pro	15	.75	14.4	–9.0	1.97	+0.6	8 [2–12]	16.0	–13.1
Claude Opus 4.7	15	.79	14.7	–11.0	1.86	+0.9	8 [4–11]	17.7	–17.0
Claude Sonnet 4.6	15	.81	15.5	–14.5	1.70	+1.7	10 [5–12]	17.8	–17.4
DeepSeek-V4-Pro	15	.79	15.6	–11.6	1.93	+1.8	10 [7–12]	18.7	–16.4
DeepSeek-V4-Flash	15	.40	16.4	–3.5	3.18	+2.6	11 [4–12]	16.0	–12.2
<i>Incomplete coverage (conditional on a valid sheet):</i>									
GPT-5.4 Mini	8	.88	10.3	–0.5	1.66	–3.2	2 [1–8]	12.4	–5.6
Llama-4 Maverick	12	.78	15.9	+9.8	4.07	+2.7	11 [5–12]	11.8	+1.0

Table 14. Study 1, fake answers by setting and judge: verdict on the negative fake answers (fail(*N*) = failing runs), positive control, and the median/p95/max score per type. Criteria in every run: repetition < 5, empty = 0, off-topic ≤ 10, reference solution ≥ 70.

Setting, judge	Battery	Pos.	Repetition	Empty	Off-topic	Model sol.
Zero-shot, Claude Opus 4.7	pass	pass	0/0/0	0/0/0	0/1/4	99/100/100
Zero-shot, Claude Sonnet 4.6	fail(6)	pass	0/100/100	0/24/81	0/0/0	100/100/100
Zero-shot, DeepSeek-V4-Pro	pass	pass	0/0/0	0/0/0	0/0/0	100/100/100
Zero-shot, GPT-5.4 Mini	fail(100)	pass	100/100/100	95/100/100	8/83/94	100/100/100
Zero-shot, GPT-5.6 Luna	fail(32)	pass	0/100/100	89/100/100	0/1/5	100/100/100
Zero-shot, Gemini 3.1 Pro	fail(20)	pass	80/100/100	100/100/100	0/0/0	100/100/100
Zero-shot, Qwen3.5-397B	fail(19)	pass	100/100/100	79/100/100	0/0/0	100/100/100
No sheet, GPT-5.4 Mini	fail(125)	fail(13)	45/69/71	5/5/7	13/21/22	82/91/95
No sheet, GPT-5.6 Luna	fail(17)	pass	0/2/93	0/0/0	8/18/19	98/100/100
Few-shot, GPT-5.4 Mini	fail(96)	–	99/100/100	95/100/100	5/89/95	–
Few-shot, GPT-5.6 Luna	fail(31)	–	0/100/100	90/100/100	0/0/0	–

Table 15. Allocation of the 170 Study 1 sheets under three routines: steps outside their fair range, sheets affected, order-inversion pairs, pairs of distinct weights that receive equal points, median relative rounding error, and steps whose points differ from those of the Study 1 routine.

Routine	Steps	Sheets	Inversions	Collapsed	Rel. error	Changed
Study 1 routine	27	11	12	458	0.033	0
Hamilton, floor first (integer)	0	0	0	286	0.033	54
Hamilton, floor first (half points)	0	0	0	0	0.020	1060

D Study 2: all preliminary results

This appendix gives the full results of the preliminary Study 2 runs that Section 6.2 summarizes: every grading basis and scoring unit (Tables 16 and 17), the fake answers (Table 18), and the generation of the sheets.

Generation. GPT-5.4 produced a valid sheet in 1 of 4 series of up to 3 attempts and GPT-5.4 Mini in 2 of 4. Of the 5 failed series, 3 ended on the cap of 10 BE per step, 1 on the rules for forks and 1 on an unreadable response. Because a generator states weights rather than BE, a step on an alternative path above the cap can only be fixed by splitting it. A valid sheet cost USD 4.15 with GPT-5.4 and USD 0.23 with GPT-5.4 Mini, failed series included.

Table 16. Expert exam: judge totals against the reference grades of the 15 D2a scripts, in points out of 100 (from the pass mark of 40 points up, 4 points make one grade point). Each value is computed per run and averaged over the runs. Grade points: the average error on the 0 to 18 scale under the exam’s key. Wrongly passed: students the judge passes whom the reference fails, and wrongly failed the reverse. Run to run: the average difference between two runs on the same script. [†] not checked with the fake answers. [‡] failed that check (Table 18).

Grading basis	Judge	Runs	Average error [95%]	Grade points	Generous (+), strict (-)	<i>r</i>	Wrongly passed / failed	Run to run
No sheet	Luna [‡]	3	14.4 [10.2, 18.5]	3.3	+12.4	.84	5 / 0	1.8
No sheet	Mini [†]	1	13.1 [8.8, 17.8]	3.0	-2.8	.39	5 / 2	-
Examiner’s sheet, per step	Luna	3	5.1 [3.3, 7.0]	1.1	-3.4	.95	1 / 0.3	2.4
Examiner’s sheet, per step	DeepSeek	1	5.2 [3.2, 7.4]	0.9	-3.9	.95	0 / 1	-
Examiner’s sheet, per step	Mini [†]	1	9.4 [6.2, 13.0]	1.7	-1.6	.74	3 / 2	-
GPT-5.4 sheet, per requirement	Luna	3	6.1 [4.2, 8.2]	1.3	+2.2	.91	2.3 / 0	4.1
GPT-5.4 sheet, per requirement	DeepSeek	1	6.2 [4.1, 8.4]	1.4	+3.1	.91	3 / 0	-
GPT-5.4 sheet, per requirement	Mini [†]	1	10.3 [6.6, 14.4]	2.3	+5.7	.74	4 / 1	-
GPT-5.4 sheet, per step	Luna [†]	3	5.7 [3.8, 8.1]	1.3	-0.4	.89	2 / 0	4.6
GPT-5.4 Mini sheet 1, per requirement	Luna	3	7.3 [4.9, 10.1]	1.6	+6.1	.92	3 / 0	3.2
GPT-5.4 Mini sheet 1, per step	Luna [†]	3	8.3 [6.4, 10.2]	1.9	+5.6	.91	4.3 / 0	3.7
GPT-5.4 Mini sheet 2, per requirement	Luna [†]	3	7.3 [4.8, 10.0]	1.7	+7.2	.94	4 / 0	3.2
GPT-5.4 Mini sheet 2, per step	Luna [†]	3	6.9 [4.2, 9.7]	1.6	+5.9	.92	4 / 0	3.2

Table 17. Expert exam, stability of GPT-5.6 Luna on the D2 scripts of each setting (no human grades needed), in points out of 100. Run to run: the standard deviation of a script’s total between runs. Between scripts: the spread of the scripts’ true totals. Dependability: the share of the spread that reflects the scripts when one run decides (0 to 1). Flips: how often a second run lands on the other side of the pass mark. A new sheet each time: one run with a freshly generated sheet, so the sheet adds its own noise, and flips there compare two sheets.

Grading basis	Scripts	Run to run	Between scripts	Dependability	Flips	US cents per run
No sheet	38	2.9	9.6	.91	0%	0.6
Examiner’s sheet, per step	38	3.4	16.2	.96	9%	1.2
GPT-5.4 sheet, per requirement	38	2.9	15.2	.97	4%	1.6
GPT-5.4 sheet, per step	38	3.6	15.2	.95	5%	1.4
GPT-5.4 Mini sheet 1, per requirement	15	3.0	14.5	.96	9%	1.2
GPT-5.4 Mini sheet 1, per step	38	3.0	13.9	.96	2%	1.0
GPT-5.4 Mini sheet 2, per requirement	15	2.6	15.7	.97	0%	1.3
GPT-5.4 Mini sheet 2, per step	38	2.7	15.2	.97	4%	1.1
A new GPT-5.4 Mini sheet each time, per requirement	15	4.1	14.8	.93	16%	-
A new GPT-5.4 Mini sheet each time, per step	38	4.0	14.3	.93	6%	-

Table 18. Fake answers and edits of the answer key on the expert exam, in points out of 100. Fake answers: the verdict on six fake answers (empty, the case text repeated, off-topic, off-topic from the same area, a keyword list, an instruction to award full marks), and max the highest score any of them received. The keyword list is built from a sheet’s terms, so it is not run without a sheet. Proposed: the most points the judge proposed for a fake answer before the quote check removed unsupported credit. Answer key: the reference solution graded as an answer. Part removed: points lost when one part of the answer key is deleted, against the points the sheet gives that part (matched by heading, a heuristic). Paragraph moved: points lost when a correct paragraph moves under another heading. Results reversed: points lost when the answer key’s results are negated.

Grading basis	Judge	Runs	Fake answers	Max	Proposed before quote check	Answer key	Part removed: lost / expected	Paragraph moved	Results reversed
No sheet	Luna	2	fail	14.0	13.0	100.0	21.0 / 0.0	2.5	37.5
Examiner’s sheet, per step	Luna	2	pass	0.0	23.0	99.5	19.5 / 21.0	1.3	18.3
Examiner’s sheet, per step	DeepSeek	1	pass	0.0	0.0	99.5	20.5 / 21.0	0.0	18.0
GPT-5.4 sheet, per requirement	Luna	2	pass	0.0	0.0	96.0	23.5 / 25.0	0.5	10.0
GPT-5.4 sheet, per requirement	DeepSeek	1	pass	0.0	0.0	94.5	21.0 / 25.0	0.0	5.0
GPT-5.4 Mini sheet 1, per requirement	Luna	2	pass	0.0	0.0	98.5	23.3 / 26.5	1.8	11.3
GPT-5.4 Mini sheet 1, per requirement	DeepSeek	1	pass	0.0	0.0	100.0	26.0 / 26.5	0.0	15.0

E Precision

This appendix shows how large a difference must be before the data can reveal it. The paper uses $\alpha = .05$ throughout, and the column for $\alpha = .025$ is given for comparison. The minimum detectable effect of a paired contrast is $(t_{1-\alpha/2, n-1} + t_{\text{power}, n-1}) \sigma_d / \sqrt{n}$. We take σ_d from the closest existing contrast, Luna with and without a generated sheet on D1, so the values are planning proxies (Table 19). The D2 proxy uses the D1 human reference, which is noisier than step-by-step expert grades; the preliminary D2a runs show smaller spreads, and Section 6.2 reports the detectable differences they imply.

Table 19. Minimum detectable effects at power .8 (t -based, paired). σ_d : SD of the paired differences in the proxy contrast.

Contrast	n	σ_d	$\alpha = .025$	$\alpha = .05$
Repeat SD, D1	45	1.87	0.88	0.80
Repeat SD, D1, 5 runs (optimistic)	45		0.63	0.57
Absolute error, D2a (15 scripts)		12.4	10.9	9.7
Absolute error, all distinct D2 scripts (38)		12.4	6.4	5.8
Absolute error, D2, dependent pair once (37)		12.4	6.5	5.9

F Fake answers (probes)

Before a model may grade, it must clear these fake answers with every sheet it will use, in each of 3 runs (Table 20), and a judge that fails a criterion in any run is not used with that sheet. E4 adds fake answers that the exam author writes, among them the reference solution rewritten along one alternative path.

Table 20. Fake answers (probes) used before any run of Study 2. Totals are out of 100 BE.

Fake answer	Input	Criterion
empty	a near-empty answer	total = 0
repetition	the case text, verbatim	total < 5
off-topic	the reference solution of another exam	total $\leq$ 10
same-area off-topic	the reference solution of another exam in the same area	total $\leq$ 10
keyword list	norms and key terms of the sheet, no sentences	total $\leq$ 10
injection	an answer that instructs the judge to award full marks	total = 0
reference solution	the exam’s own reference solution	total $\geq$ 70
section removed	the reference solution with one section removed	drop close to the removed section’s BE
paragraph moved	correct reasoning moved under another heading	drop close to 0
results reversed	result sentences of the reference solution negated	reported, no threshold

G Quote-check rules

The quote check (`services/workers/ml_evaluation/llm_judge_evaluator.py`) accepts a quote only if every fragment passes:

- *Normalization.* The text is Unicode-normalized; literal line-break escapes, markup escapes, footnote marks, emphasis markers and inline HTML are removed, and quotation marks and dashes are unified. The result marks “(+)” and “(-)” become the words *positiv* and *negativ* (positive, negative), case is folded and whitespace collapsed.
- *Fragments.* A quote is split at ellipses. Each fragment needs at least 3 tokens, or two tokens with 15 letters together, and more than a norm citation, and the whole quote needs 4 word characters.
- *Matching.* Fragments with fewer than 4 tokens must match on word boundaries. Longer ones may match as a substring or token by token in order, with up to 2 skipped answer words between quoted words and one missed quoted word per 8. Unprotected words of at least 5 letters may differ by one edit or a small inflection, and glued passages without an ellipsis are checked sentence by sentence.
- *Protected tokens.* Negations, result words and their antonyms (including every *un-* form), qualifiers such as *nur* (only) and *teilweise* (partly), Roman numerals and every token with a digit must match exactly. They may not be dropped, added, swapped or stepped over, and numbers never match part of a longer number.
- *Edges.* A match fails if a negation or a condition (*ob*, *wenn*, *falls*, *sofern*, *soweit*: whether, if, in case, provided, insofar) stands right before it in the same clause, or a negation, a negative result mark or a condition right after it in the same sentence.

H Validator checks

The validator checks every generated sheet before it is used. *Hard* checks fail an attempt and return their messages to the model. They cover:

- the document structure and every cross-reference;
- steps above 10 BE, and steps above 3 BE with a single requirement;
- requirements that can be met by an absence or that are conditional;
- steps named as a result, except the finding after a taking of evidence and the operative part of a judgment (*Tenor*);

- regular steps that depend on a step of an alternative path, and forks whose steps depend on replaced steps;
- a fork that replaces a whole section, and a path whose requirements repeat those of the replaced steps (word overlap of at least 0.6).

Soft checks only log hints. They flag broader absence patterns, forks that replace exactly one step with one step, forks above 20 BE, near-duplicate requirements, requirements that only name something and are shorter than 8 words, and a sheet on which most steps exceed 4 BE.

I Further judge rules and designs for alternative paths

Besides the rules of Section 4.3, the judge follows two more:

- *Unforeseen paths*. If the answer takes a defensible path that the sheet does not foresee, the judge credits the corresponding step at equal depth of reasoning and flags the script for human review.
- *General rules*. Contradictory results without a decision fail the requirement on the result, and unnecessary alternative analyses earn nothing. Length, many citations and strings of keywords are not performance, whereas a short, correct application meets a requirement fully.

A dispute that changes only a result or its reasons (a local alternative) stays inside a step as an outcome-neutral requirement, such as “discusses the dispute with the arguments of both sides, both results defensible”. A structural fork replaces named steps of the reference solution’s path with the steps of each alternative path, within the same budget. E4 compares three designs:

- *A0, replace*. Only the reference solution’s path is scored, and the rendered sheet tells the judge to credit work on another defensible path on the replaced steps. Study 1 used this design, which needs the step unit, because a requirement of a replaced step describes that step’s own performance.
- *A1, branch, best path*. The judge scores the steps of every path, and code counts the path with the most points.
- *A2, branch, declared path*. As A1, but code counts the path the judge declares the answer follows, backed by a verified quote; without one it counts the reference solution’s path.

A1 and A2 come from the same call. A path counts as followed only if the answer argues it as its defensibility requirements demand, so a bare other result does not count.

J Data statement and costs

D1. The Study 1 corpus of 15 German exams and 45 answers with blind human grades [50]. Exam texts are third-party material used under the permissions documented for that corpus and are not redistributed, and human grades are pseudonymized.

D2. One police-law exam with its author’s sheet and key, 15 graded scripts (D2a) and 23 student submissions to a practice exam (D2b), used with the students’ consent for research. All texts are pseudonymized before any research call to a model provider: the check found and removed 1 line with a student’s name and 1 examiner header, and 9 grade spreadsheets carried a private e-mail address, which no derived file contains. The material stays private, and only aggregate statistics are published. [PENDING: who wrote the scripts and when, exam status, consent process, storage location and deletion date]

Z2. Gregor Roth’s grading sheet and workbook for a civil-law exam, without case, solution or scripts, which stay private and are described only by their structure.

Model calls. Every call records the model id, parameters, prompt hashes and the code version. For D1 we release aggregates, since the per-call records contain third-party exam text. [PENDING: owner: release the per-call records with the dataset release?]

Table 21. Planned cost in USD per judge call (about 25k input and 6k output tokens) and per generated sheet, from measured tokens at catalog prices. Open-model sheets cost less than the value shown. [PENDING: measured costs]

Model	USD
GPT-5.6 Luna	0.013
GPT-5.4 Mini	0.046
DeepSeek-V4-Pro	0.070
Gemini 3.1 Pro	0.120
GPT-5.4	0.150
Claude Sonnet 4.6	0.170
Sheet: open models	0.05
Sheet: GPT-5.4 Mini	0.15
Sheet: GPT-5.4	0.60
Sheet: Claude Opus 4.7	1.00

Costs. Table 21 lists the planned cost per judge call and per generated sheet; the program is planned at USD 140 to 200 (Table 2).

K Output specification of the checklist sheet

[PENDING: the verbatim output specification (German) and its English translation, after the E0 freeze]

L Example generated sheet

[PENDING: one generated checklist sheet for a D1 exam, as rendered for the judge, after the E0 freeze; D1 only, never D2]